PRD - Retail Agentic AI Platform – Demand, Inventory, Pricing & Brand Intelligence

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1. Business Problem & Market Context

1.1 Business Problem

Retail and apparel enterprises operate in an environment where:

- Consumer demand is **highly volatile** and trend-driven
- Planning cycles are **slow, periodic, and reactive**
- Decision-making is fragmented across demand planning, inventory, pricing, production, and marketing
- Most tools generate **reports**, not **decisions**

This results in:

- Excess inventory → markdowns → margin erosion
- Stockouts → lost revenue and poor customer experience
- Late reaction to social and fashion trends
- Overproduction and sustainability risk
- Heavy dependency on manual planners and Excel-based workflows

Core Problem Statement

> Retail organizations lack a unified system that can continuously convert real-time signals into coordinated, explainable, and executable business decisions.

1.2 Market Context

- ERP / WMS / OMS are **systems of record**, not decision engines
- BI tools answer *"what happened"*, not *"what should we do now"*
- Point AI tools (forecasting, pricing, allocation) operate in silos
- Fashion cycles have compressed to weeks due to social media
- Enterprises are increasingly exploring **Agentic AI** to:
    - Reduce decision latency
    - Handle cross-functional trade-offs
    - Maintain human control and governance

2. Customer & User Needs

Business Needs

- Faster reaction to demand and trend shifts
- Reduced excess inventory and markdown dependency
- Higher full-price sell-through
- Scalable planning independent of individual planners
- Measurable sustainability impact
- Enterprise-grade governance and auditability

User Needs

- Clear, explainable recommendations
- Ability to approve, override, or auto-execute decisions
- Visibility into confidence, trade-offs, and impact
- One command center across all planning functions

3. Product Objectives

Primary Objectives

1. Turn planning from static reports into executable decisions
2. Continuously align supply, pricing, and campaigns with demand
3. Reduce waste, markdowns, and firefighting
4. Enable human + AI collaborative decision-making

Secondary Objectives

- Improve planner productivity
- Codify institutional knowledge
- Support sustainability and ESG goals

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4. Solution Overview

The Retail Agentic AI platform is a **multi-agent decision intelligence system** that:

1. **Senses** signals (sales, trends, social, weather, inventory)
2. **Reasons** via specialized AI agents
3. **Orchestrates** decisions across functions
4. **Executes** actions with configurable human approval
5. **Explains** every recommendation
6. **Learns** from outcomes and human feedback

The product is delivered as a **central AI Command Center**, as reflected in the UI screens:

- Dashboard
- Agent Insights
- Trends & Campaigns
- Decision Queue
- Settings & Governance

5. User Personas & How they will use the tool

5.1 CXO (CEO / COO / CFO)

Goals

- Margin protection
- Working capital efficiency
- Risk reduction
- Strategic visibility

Usage

- Executive dashboard view of:
    - Inventory health
    - Demand vs supply gaps
    - Financial and sustainability impact

- Configure approval thresholds (via Settings → Agent Config)
- Review high-impact decisions in Decision Queue

5.2 Marketing & Brand Teams

Goals

- Successful launches and campaigns
- Trend-aligned promotions
- Inventory readiness

Usage

- Use **Trends & Campaigns** module:
  - Track emerging trends (momentum, sentiment, growth)
  - Link trends to SKUs
- Use **Brand Campaign Agent** to:
  - Predict campaign uplift
  - Generate creatives (banner, social, email, WhatsApp)
  - Validate supply readiness before launch

5.3 Merchandising / Product Development

Goals

- Right product, right quantity
- Reduced overproduction
- Faster trend response

Usage

- Review demand forecasts and trend intelligence
- Evaluate production batch recommendations
- Assess sustainability score, waste estimate, and overproduction risk
- Approve or adjust production plans

5.4 Supply Chain & Planning Teams

Goals

- Optimal inventory placement
- Reduced stockouts and overstocks
- Fewer manual interventions

Usage

- Review inventory optimizer recommendations:
  - Replenish
  - Redistribute
  - Markdown
- Monitor store-level demand maps
- Execute approved inventory movements

6. Value Proposition to the Business

6.1 Turns Planning From Reports Into Decisions That Execute

- Moves from dashboards to **actionable, system-driven decisions**
- Connects insights directly to execution paths
- Reduces insight-to-action latency from weeks to hours

6.2 Handles Fashion Volatility Humans Can't

- Continuously monitors trends and sentiment
- Reacts faster than human planning cycles
- Adapts pricing, inventory, and production in near real time

6.3 Aligns the Entire Value Chain

- Resolves trade-offs across margin, availability, and sustainability
- Prevents siloed optimization
- Optimizes enterprise-level outcomes

7. Functional Requirements

7.1 Agent Framework (Common)

Each agent must expose:

- Status (Active / Processing / Idle)
- Confidence score
- Active recommendations
- Last run timestamp
- Explainability metadata

7.2 Agent Details

1. Demand Intelligence Agent

Purpose
Predict near-real-time and future demand at SKU × Region × Channel level.

Inputs

- Historical sales & sell-through
- Trend momentum signals
- Weather & seasonality
- Campaign & promotion calendar

Core Capabilities

- Multi-horizon demand forecasting (daily / weekly / seasonal)
- Demand volatility & anomaly detection
- Scenario simulation (best / base / worst case)
- Forecast confidence scoring
- Demand signal fusion across structured + unstructured data

Outputs

- Demand forecasts (SKU × Region × Time)

- Volatility & confidence scores
- Demand risk alerts

2. Trend Watcher Agent

Purpose
Detect emerging fashion, lifestyle, and cultural trends early.

Inputs

- Social media (Instagram, Twitter, FB, Youtube)
- Search trends
- Fashion blogs & news
- Influencer content

Core Capabilities

- Trend detection & clustering
- Momentum scoring (velocity + acceleration)
- Sentiment analysis
- Regional trend differentiation
- Trend-to-category / SKU mapping

Outputs

- Active & emerging trend list
- Momentum & sentiment scores
- Linked SKUs & categories

3. Inventory Optimization Agent

Purpose
Continuously optimize inventory levels and placement.

Inputs

- Demand forecasts
- Inventory on hand & in transit

- Sell-through velocity
- Store/DC constraints

Core Capabilities

- Replenishment quantity optimization
- Inter-store / inter-DC redistribution
- Stockout & overstock risk prediction
- Markdown trigger identification
- Priority scoring (high / medium / low)

Outputs

- Replenishment recommendations
- Redistribution plans
- Inventory risk alerts
- Markdown suggestions

4. Pricing & Promotion Agent

Purpose
Maximize revenue and margin through intelligent pricing.

Inputs

- Demand elasticity
- Inventory pressure
- Competitive pricing
- Seasonality & lifecycle stage

Core Capabilities

- Price elasticity modeling
- Optimal price point recommendation
- Markdown timing optimization
- Revenue & margin impact simulation
- Guardrail enforcement (max price change %)

Outputs

- Price change recommendations
- Expected revenue & margin lift
- Confidence score per recommendation

5. Production & Sustainability Agent

Purpose
Align production volumes with demand while minimizing waste.

Inputs

- Aggregated demand forecasts
- Inventory health
- Manufacturing capacity & lead times
- Sustainability constraints

Core Capabilities

- Production batch size optimization
- Overproduction risk prediction
- Waste & fabric utilization estimation
- Sustainability impact scoring
- Batch-splitting recommendations

Outputs

- Production plans
- Waste & sustainability scores
- Overproduction risk alerts

6. Store Replenishment Agent

Purpose
Optimize store-level inventory availability.

Inputs

- Store-level demand patterns
- Regional trends

- Store capacity constraints

Core Capabilities

- Store-specific demand forecasting
- Shipment prioritization
- Critical store detection
- Regional demand imbalance analysis
- Service-level optimization

Outputs

- Store shipment plans
- Critical store alerts
- Regional demand insights

7. Brand Campaign Agent

Purpose
Ensure campaigns and launches are demand-aligned and supply-ready.

Inputs

- Trend signals
- Inventory availability
- Historical campaign performance
- Pricing & promotion plans

Core Capabilities

- Campaign demand uplift prediction
- Launch timing optimization
- Inventory readiness validation
- Creative generation (banner, social, email, WhatsApp)
- Campaign ROI simulation

Outputs

- Campaign recommendations

- Generated creatives
- Expected uplift & ROI

7.3 Orchestration Layer

Central Orchestrator Responsibilities

- Coordinate agent outputs
- Resolve conflicts (e.g., pricing vs sustainability)
- Decide execution mode:
    - Auto-execute
    - Human approval
- Maintain decision audit trail

7.4 Human-in-the-Loop & Governance

Configured via **Settings → Agent Config**

Approval Thresholds

- Price changes beyond **X%**
- Production volume changes beyond **Y%**
- Inventory transfers beyond **Z units/value**

Governance Features

- Confidence-based auto-execution
- Sustainability checks toggle
- Decision expiry
- Override and feedback capture

7.5 Explainability Layer

For every decision, the system must display:

- Why the agent recommended this
- Input signals used
- Confidence score

- Expected impact (revenue, margin)
- Trade-offs considered

8. Non-Functional Requirements

- Role-based access control
- Enterprise-grade audit logs
- Modular agent architecture
- API-first integration
- Scalable to millions of SKUs

9. Out of Scope (MVP)

- Fully autonomous execution without guardrails
- Custom per-client model training

10. Success Metrics & Business Metrics (NEW)

These are the **primary metrics leadership will track** to measure platform success.

Full-Price Sell-Through (%)

Indicates improved demand alignment and reduced dependency on markdowns.

- **Demand Intelligence Agent**
  - Prevents overproduction
- **Trend Watcher**
  - Catches trends early
- **Inventory Optimizer**
  - Moves stock before it becomes stale
- **Pricing Agent**
  - Avoids unnecessary early markdowns
- **Brand Campaign Agent**
  - Pushes demand while stock is still fresh

Excess Inventory Reduction (%)

Measures effectiveness of demand, inventory, and production agents in reducing overstock.

Inventory Turnover Improvement

Tracks working capital efficiency and supply chain velocity.

Revenue / Margin Uplift from AI Decisions

Captures direct financial impact from:

- Pricing optimization
- Inventory redistribution
- Campaign alignment

Decision Velocity (Insight → Action Time)

Measures how fast the organization moves from signal detection to execution (e.g., days → hours).